

So, in public transport systems such as buses that have to stop frequently at known points where charging facilities are installed, using supercapacitor as the energy storage device is quite viable.

Since 2010, capa buses have been operating in Belgrade in Serbia, Shanghai, Hong Kong, Sofia, Bulgaria, and Graz in Austria without any major trouble. These buses are provided with supercapacitors made of activated carbon and have a typical energy density of about 10 watt-hours/kg compared to 200 watt-hours/kg of a high-performance Li-ion battery. These buses consume energy that is only one-tenth of the cost of fuel consumed by a diesel bus and can achieve lifetime fuel savings of about ₹ 15 million per bus.

The buses use 40 per cent less electricity, even when compared to an electric trolley bus, mainly because they are lighter. The capa bus is 40 per cent cheaper than a Li-ion battery bus and far more reliable. The buses need to stop regularly every 5km allowing

quick recharging at charging stations at bus stops. A collector on top of the bus rises a few feet and touches an overhead charging line at the stop. Within a few minutes, the supercapacitor banks stored under the bus seats are fully charged. The buses also capture energy from braking.

The recharging stations can be equipped with solar panels. A third-generation bus has been developed, which can extend the range up to 32km. In the future, buses would use supercapacitors with vertically aligned carbon nanotubes as the active element that would have one-quarter of the energy density of a lithium-ion battery. Future developments include the use of inductive charging under the street to avoid overhead wiring.

Spot welding gun. Spot welding guns are generally used in the current robotic system in assembly lines for resistance welding of thin gauge steel (about 0.8mm). These machines use a pneumatic/electromechanically operated forcing system to apply force to electrodes and power supply to provide energy for welding. It weighs about 50kg including the weight of the transformer, which is about 25kg.

These welding guns are gradually replaced with supercapacitors based devices that result in a reduction in weight by 50 per cent due to elimination of bulky transformers. It yields a reduction in the cost of

₹ 7 million per unit and also reduces the scale of automation in the assembly line. These new generation machines are provided with supercapacitors up to a level of 5000 F and operate at a voltage less than 10V.

Defence application

Where to use supercapacitors in military devices? In applications where the charge can more or less be maintained continuously, but at times it is required to obtain enormous amounts of energy impulsively in a very short period of time. Supercapacitors already have set their foot in military technology.

Laser weapons (LaWs). Laser weapons no longer exist only in science fiction. USA, Russia, China, and India have systems with which enemy aircraft, ballistic missiles, anti-ship ballistic missiles are destroyed with a laser beam. Onboard generator of the aircraft or ship provides electrical energy that is stored in the supercapacitor banks.

In case of a fast moving distant target like an enemy aircraft, missile, etc, it is not possible to hold the laser beam on the target. Therefore the energy that can destroy the target must be delivered in the form of single radiation of 10-100kW power pulse in a few milliseconds. Only supercapacitors have the capability of doing this task. Laser weapon systems are tied to ship or aircraft's navigation radar and can be operated through the mother vehicle's combat information centre.

A typical arrangement of LaWs is shown in Fig. 6.

Railgun. A railgun is a military device that uses electromagnetic force known as Lorentz force to launch projectiles 2m long and weigh 100-120kg at speeds exceeding Mach 7-8 using a sliding armature that is accelerated along with a pair of conductive rails. The projectile normally does not contain explosives, but relies on the projectile's high speed to inflict damage.

The railgun uses a pair of parallel conductors, or rails, along which a sliding armature is accelerated by the electromagnetic effects of a current that flows down on one rail



Fig. 6: Laser gun fitted in a destroyer

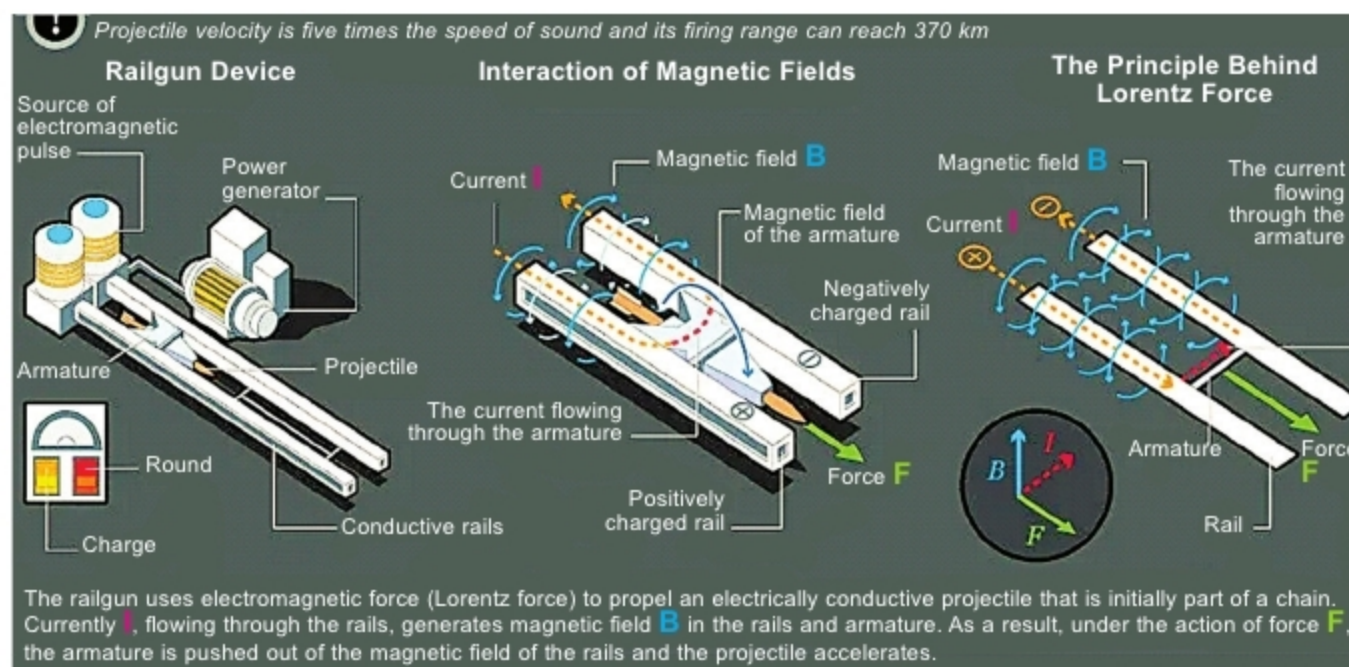


Fig. 7: Railgun